

ALVE IBRAHIM

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½ Uttar Badda, Dhaka

EDUCATION

BRAC University Dhaka, Bangladesh *B.Sc. in Computer Science and Engineering Jun 2021 – Dec 2025 –*

Relevant Coursework: Software Engineering, System Analysis & Design, Machine learning, Android App Development, Data Structures & Algorithms.

PROJECTS

[Azure Business Solutions](#) | *React, Vite, TypeScript, Convex, Three.js, Tailwind CSS, Framer Motion, Locomotive Scroll* – Developed as a fully custom client project using React 19, Vite 6.2, TypeScript 5.7 for responsive and performant UI.

- Convex 1.21 with real-time database and authentication (including anonymous and password login). –
- Tailwind CSS 3 for utility-first styling, Framer Motion 12 and Three.js 0.176 for smooth animations, 3D visuals, and immersive scroll effects via Locomotive Scroll 4.1.
- Contact & lead capture via Web3Forms and email integration.

[Artefactum Website](#) | *React, Vite, Chakra UI, Cloudinary, Tailwind CSS*

- Developed as a fully custom client project, tailored to their branding and workflow.
- Built with React 19 and Vite for fast, optimized performance.
- Styled using Tailwind CSS and Chakra UI for a clean, responsive UI.
- Cloudinary integration for image hosting, optimization, and fast CDN delivery.
- Includes gallery system, artist portfolio sections, and mobile-first layout, integrated with Artwork Archive for product and artwork database management.

PC Builder BD | *Flutter, Supabase, PostgreSQL*

- Developed a cross-platform PC build recommendation app using Flutter, delivering tailored component suggestions based on budget and user needs.
- Integrated Supabase for authentication, secure user sessions, and real-time database storage. –
- Implemented PostgreSQL for scalable, relational data management of components and user preferences.

Study-Sync Pro | *JavaScript, Supabase, React Native/EAS*

- Built a collaborative study app with JavaScript and React Native, enabling real-time group study sessions. – Deployed backend on Supabase for database, authentication, and session syncing.
- Ensured cross-platform usability by supporting both web and mobile through EAS (Expo Application Services). – Developed polished interfaces and streamlined interactions to facilitate virtual study collaboration.

Pothole Detection Model | *Python3, YOLOv8*

- Built an instance segmentation model using YOLOv8 to detect potholes on custom

datasets. **ACADEMIC THESIS**

Advanced Personnel Security System Using Behaviour and Activity Analysis Oct 2025 – Proposed an AI-driven multi-modal surveillance framework to detect insider threats in high-security facilities (e.g., nuclear power plants).

- Utilized Transformer-based pose encoders, 3D CNNs, and YOLOv8 for body-language, emotion, and weapon detection from surveillance feeds.
- Applied SVM, Random Forest, and Decision Trees for keystroke and anomaly analysis, and BERT-based sentiment models for social media threat monitoring.
- Integrated all modules into a unified personnel risk assessment and alert system compliant with IAEA security standards.
- Achieved high detection accuracy across multimodal datasets, enhancing proactive threat identification in critical infrastructures.

TECHNICAL SKILLS

Languages: Python, C, JavaScript, TypeScript, HTML, CSS

Web Development: ReactJS, Django, MERN Stack, NextJS, Shopify

Databases: MySQL, MongoDB, PostgreSQL

Version Control: Git, GitHub

Productivity Tools: Microsoft Excel, Word, PowerPoint